PAVAN KUMAR MATHI

Embedded Systems & Hardware Engineer | Embedded C · ARM7 · IoT · Robotics

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SUMMARY

Final-year B.Tech ECE student (2027) focused on embedded systems and robotics. Wrote register-level Embedded C firmware for four ARM7 LPC2148 systems (RTC, timers, interrupts, ADC, UART, I2C, SPI, watchdog), is building an Arduino UNO Q rover, and completed internships in embedded IoT and PCB design. Seeking an entry-level embedded, IoT or core electronics role.

TECHNICAL SKILLS

Microcontrollers: ARM7 LPC2148, Arduino UNO / UNO Q, ESP32, ESP32-CAM

Embedded: Embedded C, register-level programming, interrupts (VIC), timers and PWM, RTC, ADC, watchdog, bootloader flashing

Interfaces: UART, I2C, SPI, PS/2, 16x2 LCD, 74HC595, AT24C256 EEPROM, motor drivers, LM35 / LDR / rain / soil-moisture sensors

Hardware & Tools: PCB and circuit design, power budgeting, Keil uVision4, Arduino IDE, arduino-cli, Fusion 360, AutoCAD, Blynk, Git, Linux

Programming: C, Python (OpenCV, YOLOv8 for robot vision)

INTERNSHIPS

Embedded Systems & IoT Intern – Sreshta Semicon Technologies Pvt. Ltd. **Jun 2026 – Jul 2026**

- Hands-on training and project work in embedded development and IoT architecture, connecting sensor data from hardware to software systems.

PCB Design & Electronics Intern – Pantech Solutions **May 2026 – Jun 2026**

- PCB layout workflow, circuit design practice and 8255-based microcontroller board fundamentals.

PROJECTS

A.I.R – Artificial Intelligence Rover (Final-Year Project) | *Arduino UNO Q, Embedded C, Python, YOLOv8*

- Designing an offline companion rover on the Arduino UNO Q: MCU firmware for motor control, sensors and emergency stop (17% of flash) talks to a Linux side that runs vision and voice.
- Prepared the pin map, power design and bill of materials (Rs 21k, costed with live prices); built a web test panel to bring up each peripheral before final assembly.
- Trained a YOLOv8n detector (mAP@50 0.951) and tested the full AI-to-MCU control path in a simulator with 0 wall collisions over a 5-minute run.

Digital Countdown Timer with RTC – ARM7 LPC2148 (team of 3) | *Embedded C, Keil uVision4*

- Clock, calendar, 12/24 h, alarm with snooze, hourly chime, stopwatch with laps and a potentiometer-set countdown (ADC), on a 1 ms Timer0 interrupt base.
- Traced random board resets to spurious IRQs and fixed them with a default ISR and larger stacks; verified on hardware.

LPC2148 Security & Monitoring Systems (3 projects) | *Embedded C, I2C, UART, SPI, Watchdog*

- Secure Access & Attendance: PS/2 ID + PIN entry, user records and attendance log in AT24C256 I2C EEPROM, RTC time stamps, LM35 temperature (ADC), UART reports to PC, Timer1 lockout; builds with 0 errors, 0 warnings.
- Smart Security lock (30 s lockout, access log, watchdog reset) and a Multi-Peripheral monitor driving 6 LCD screens, 74HC595 LEDs over SPI and a 1 Hz hardware square wave on MAT1.2.

ESP32-CAM Surveillance Robot and Weather Monitoring IoT Node | *ESP32-CAM, Motor Drivers, Sensors, Blynk*

- Wi-Fi robot car with MJPEG live video, browser control and PWM motor-speed control; adding on-board human detection.
- Community IoT node (team of 4) streaming LDR, rain and soil-moisture readings to a Blynk dashboard.

AI Medicine-Delivery Drone – Design Support | *INAV 9, Power Budgeting*

- Selected components and calculated thrust (2.09:1 thrust-to-weight), hover current and endurance for 300 g and 500 g payload quadcopters; prepared the INAV flight-controller setup.

EDUCATION

B.Tech, Electronics and Communication Engineering – MVGR College of Engineering (A), Vizianagaram **2023 – 2027**

Intermediate (MPC) – Sri Chaitanya Junior College, Vizianagaram **2021 – 2023**

ACHIEVEMENTS & LEADERSHIP

- Team Lead – ranked 28th of 100+ teams, Autodesk 36-hour hackathon (ANITS, Jan 2026); built the prototype's Fusion 360 CAD.
- SPOC Coordinator – organised an electronics workshop at MVGR's Aadhrita tech fest and mentored 50+ participants.

Languages: English, Telugu, Hindi